Experiment 5-A

Reports for the Transport Management System

Aim

To design and build a complete suite of Reports in Zoho Creator for the Transport Management System, demonstrating all nine Report Types: List, Spreadsheet, Kanban, Calendar, Map, Timeline, Pivot Table, Chart and Summary View.

Procedure

1. The coordinator schedules trips through the Trip_Scheduling form using lookup fields.
2. The coordinator monitors upcoming trips using All_Trips_List and Trip_Activity_Timeline.
3. Drivers and coordinators track trip progress visually using Trip_Status_Kanban.
4. The coordinator uses Trip_Schedule_Calendar to plan vehicle and driver allocation per day.
5. The route planner uses Route_Location_Map to view route starting points and plan coverage.
6. The manager reviews VehicleType_By_TripStatus_Pivot and RouteWise_Trip_Chart to plan fleet use.
7. Daily, the coordinator updates trips in bulk with Trip_Schedule_Spreadsheet, while the manager uses VehicleType_Wise_Summary to review fleet capacity.

Reports Used

Report Type	Report Name	Purpose and Key Configuration
List View	All_Trips_List	Flat searchable list; columns Trip ID, Trip Date, Vehicle, Driver, Route, Departure Time, Trip Status; sorted by Trip Date ascending
Spreadsheet View	Trip_Schedule_Spreadsheet	Bulk editing; only Trip Status is editable; grouped by Route
Kanban View	Trip_Status_Kanban	Cards grouped by Trip Status: Scheduled / In Progress / Completed / Cancelled
Calendar View	Trip_Schedule_Calendar	Based on Trip_Date; trips per day; click a date to drill down
Map View	Route_Location_Map	Starting_Point of Route_Details (geocode-enabled address field); pins show route locations
Timeline View	Trip_Activity_Timeline	Based on Trip_Date; grouped by week to show trip demand trend
Pivot Table	VehicleType_By_TripStatus_Pivot	Rows Vehicle_Type (from the Vehicle lookup), columns Trip_Status, values count of trips; blanks shown as zero
Chart View	RouteWise_Trip_Chart	Pie chart of trips per Route; auto-refresh on new trip
Summary View	VehicleType_Wise_Summary	Grouped by Vehicle_Type; count of trips and average Seating_Capacity; zero-count types still listed

Validation & Configuration Rules

- List View sorted by Trip Date ascending so the earliest trips appear first.
- Spreadsheet View edit rights restricted to the Transport Coordinator role; all other columns locked.

- Kanban card movement updates the Trip Status field and triggers the vehicle status workflows.
- Calendar View shows trips by Trip Date; click a date to see that day's trips.
- Map View requires the Starting Point field to be geocode-enabled.
- Timeline View groups trips by week using Trip Date.

Output Screenshots

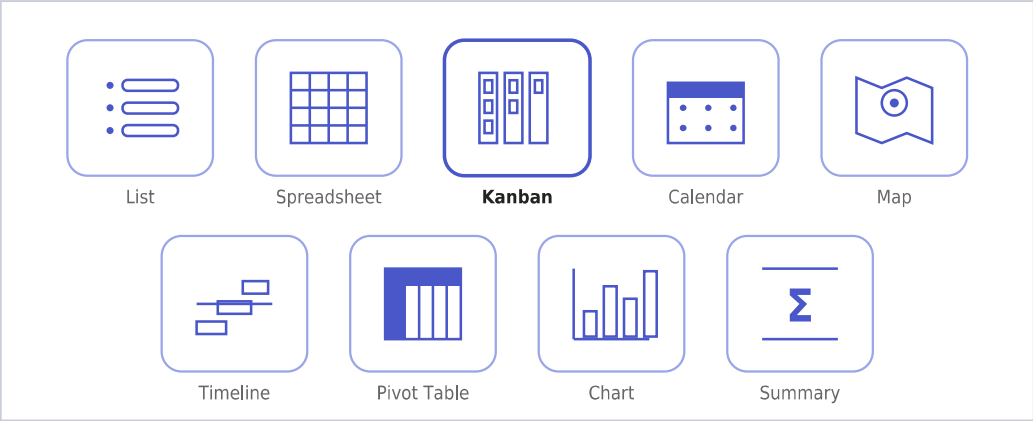


Figure 1: Report type selector in Zoho Creator's Report Builder showing the nine report types used (illustrative mock-up, not a screenshot).

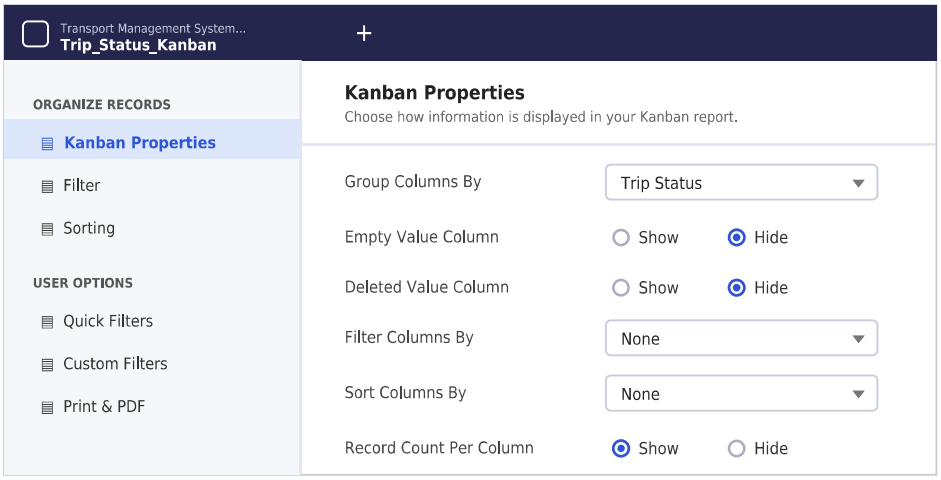


Figure 2: Kanban Properties of Trip_Status_Kanban: columns grouped by Trip Status (illustrative mock-up).

Sample Data Used by the Reports

Sample data for this lab record. Values shown are the expected output for these sample records.

Trip ID	Vehicle Number	Vehicle Type	Seats	Driver	Route	Trip Status
TRP-001	TN 01 AB 1234	Bus	40	Arun Kumar	Chennai – Vellore	In Progress
TRP-002	TN 02 CD 5678	Van	12	Suresh Babu	Chennai – Kanchipuram	Scheduled
TRP-003	TN 05 JK 7890	Mini Bus	25	Mohan Raj	Chennai – Pondicherry	Completed
TRP-004	TN 03 EF 9012	Car	4	Karthik S	Chennai – Vellore	Completed
TRP-005	TN 04 GH 3456	Bus	52	Ravi Chandran	Chennai – Kanchipuram	Cancelled

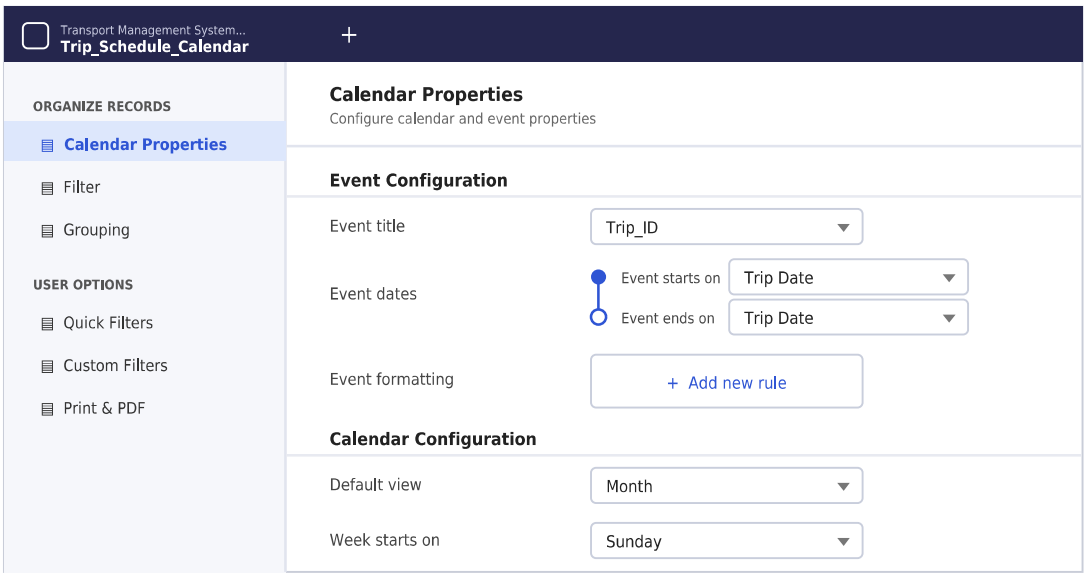


Figure 3: Calendar Properties of Trip_Schedule_Calendar: Trip_ID as event title, Trip Date as the event date (illustrative mock-up).

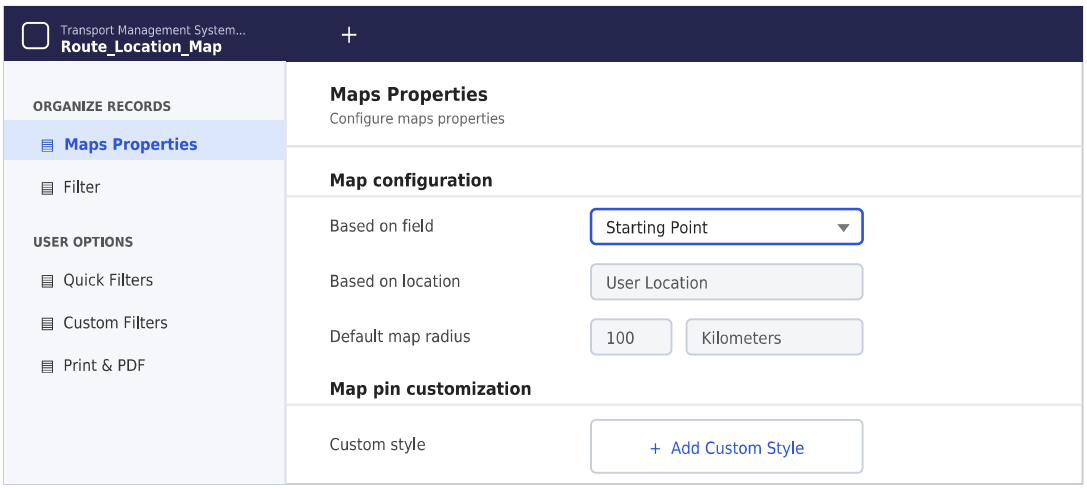


Figure 4: Maps Properties of Route_Location_Map: based on the Starting Point field (illustrative mock-up).

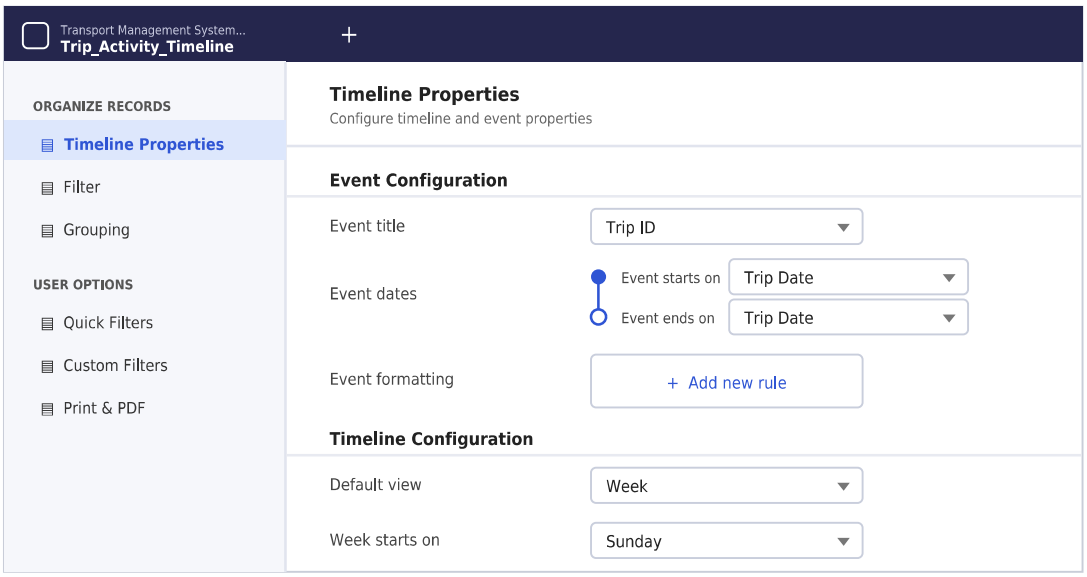


Figure 5: Timeline Properties of Trip_Activity_Timeline: Trip ID as title, Trip Date as the event date, default Week view (illustrative mock-up).

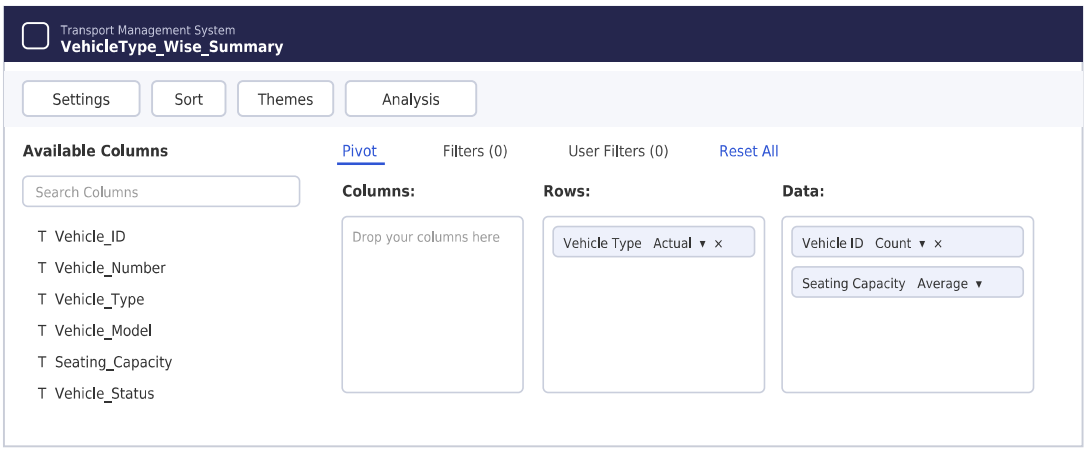


Figure 6: Summary view builder: VehicleType_Wise_Summary with Vehicle Type grouped in rows (illustrative mock-up).

Expected Pivot Table: VehicleType_By_TripStatus_Pivot

Vehicle Type	Scheduled	In Progress	Completed	Cancelled	Total
Bus	0	1	0	1	2
Mini Bus	0	0	1	0	1
Van	1	0	0	0	1
Car	0	0	1	0	1
Truck	0	0	0	0	0

Blank cells are displayed as zero, as configured.

Expected Summary: VehicleType_Wise_Summary

Vehicle Type	Count of Trips	Average Seating Capacity
Bus	2	46
Mini Bus	1	25
Van	1	12
Car	1	4
Truck	0	0

Expected Chart and Kanban Counts

RouteWise_Trip_Chart (pie)	Trips	Kanban column (Trip Status)	Cards
Chennai – Vellore	2	Scheduled	1
Chennai – Kanchipuram	2	In Progress	1
Chennai – Pondicherry	1	Completed	2
		Cancelled	1

Result

A suite of nine Reports, one for each Report Type, was designed on the Vehicle_Details, Route_Details and Trip_Scheduling forms of the Transport Management System in Zoho Creator, giving each stakeholder a purpose-built view of transport data.